**Project Design Phase-II**
**Technology Stack (Architecture & Stack)**

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|---|---|
| Date | 25 June 2026 |
| Team ID | SWTID-2026-5525 |
| Project Name | Visualization Tool for Electric Vehicle Charge and Range Analysis |
| Maximum Marks | 4 Marks |

**Technical Architecture:**

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

**Example: Order processing during pandemics for offline mode**

**Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/**

**User**
↓
**Flask Web Application**
↓
**Tableau Embedded Dashboard**
↓
**Tableau Public Server**
↓
**Tableau Desktop**
↓
**MySQL Database Server**
↓
**EV CSV Datasets**

Guidelines:

Include all the processes (As an application logic / Technology Block)
Provide infrastructural demarcation (Local / Cloud)
Indicate external interfaces (third party API's etc.)
Indicate Data Storage components / services
Indicate interface to machine learning models (if applicable)

**Datasets:**
**- ElectricCarData**
**- EVIndia**
**- Charging Stations**
**- Cheapest EV Cars**

**Table-1 : Components & Technologies:**

| S.No | Component | Description | Technology |
|---|---|---|---|
| 1 | User Interface | Web page to display EV analytics dashboard | HTML, CSS, JavaScript |
| 2 | Application Logic-1 | Backend web application integration | Python Flask |
| 3 | Application Logic-2 | Dashboard creation and data visualization | Tableau Desktop |
| 4 | Application Logic-3 | Data processing and queries | SQL |
| 5 | Database | Stores EV datasets and structured information | MySQL Server 8.0 |
| 6 | Cloud Database | Published dashboard hosting platform | Tableau Public |
| 7 | File Storage | Stores project files and datasets | Local File System, GitHub |
| 8 | External API-1 | Dashboard embedding service | Tableau Embed API |
| 9 | External API-2 | Not Applicable | Not Applicable |
| 10 | Machine Learning Model | Not Applicable | Not Applicable |
| 11 | Infrastructure | Application execution and deployment | Local Server, Flask Server |

**Table-2: Application Characteristics:**

| S.No | Characteristics | Description | Technology |
|---|---|---|---|
| 1 | Open-Source Frameworks | Frameworks used for web integration and development | Flask, Python |
| 2 | Security Implementations | Secure database connection and dashboard access management | MySQL Authentication, Tableau Public Access Control |
| 3 | Scalable Architecture | Separate database, visualization and web layers | Three-Tier Architecture |
| 4 | Availability | Dashboard can be accessed online anytime | Tableau Public |
| 5 | Performance | Optimized queries and interactive dashboard loading | SQL Optimization, Tableau Extract |

**References:**


https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d